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AUGUST 1 1918 SERIAL NO. 160

THE MENTOR

PHOTOGRAPHY

By

PAUL L. ANDERSON

DEPARTMENT OF FINE ARTS

VOLUME 6 NUMBER 12

TWENTY CENTS A COPY

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A Voice From Far Cathay

Dear Mentor:—One of my most respected college professors advised his classes to review their several groups of studies every seven years, and in the broad, I agree with the advice. It is just this review that The Mentor gives some of us. The Mentor is not *learned*. It furnishes a most readable review, with pithy editorials and discriminatively selected pictures. It can be appreciated by the man who has never been outside the town of his birth, and it can be enjoyed by the person who has converted stamped gold into the legal tender which lets one into the city, or gallery, or park, or museum, or observatory described; and it can be read with profit by the one who is interpreting life in the class room.

I usually read it by bits between courses at the dinner table, and it often has taken the place of another seat. It has this advantage: it never talks shop, save in an entertaining way.

The pictures make good material for "identification" or "information tests." A selection of twenty makes good material for one "stunt" of an evening for a small group of guests.

But I like them best for use in a bulletin board in my class room. With titles or brief notes translated into Chinese, they attract the students around the board between classes. It is an easy matter to have a series of fifteen or twenty groups through the year, that are of interest to one's students, and give real information and stimulus.

This letter is not designed to lead you to believe that the publication takes its place with the essential possessions of the American missionary in the Orient—the Bible, Montgomery Ward catalog, and tennis racquet—but it is written that you may know that it helps one to keep "fit."

In appreciation, yours,

DANIEL S. DYE

West China Union University
Chengtu, West China

THE MENTOR ASSOCIATION

ESTABLISHED FOR THE DEVELOPMENT OF A POPULAR INTEREST IN ART, LITERATURE, SCIENCE, HISTORY, NATURE, AND TRAVEL

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AUGUST 1st, 1918. VOLUME 6. NUMBER 12.

BY PAUL L. ANDERSON

PHOTOGRAPH FROM A DAGUERREOTYPE

PHOTOGRAPHY

A Daguerreotype

ONE

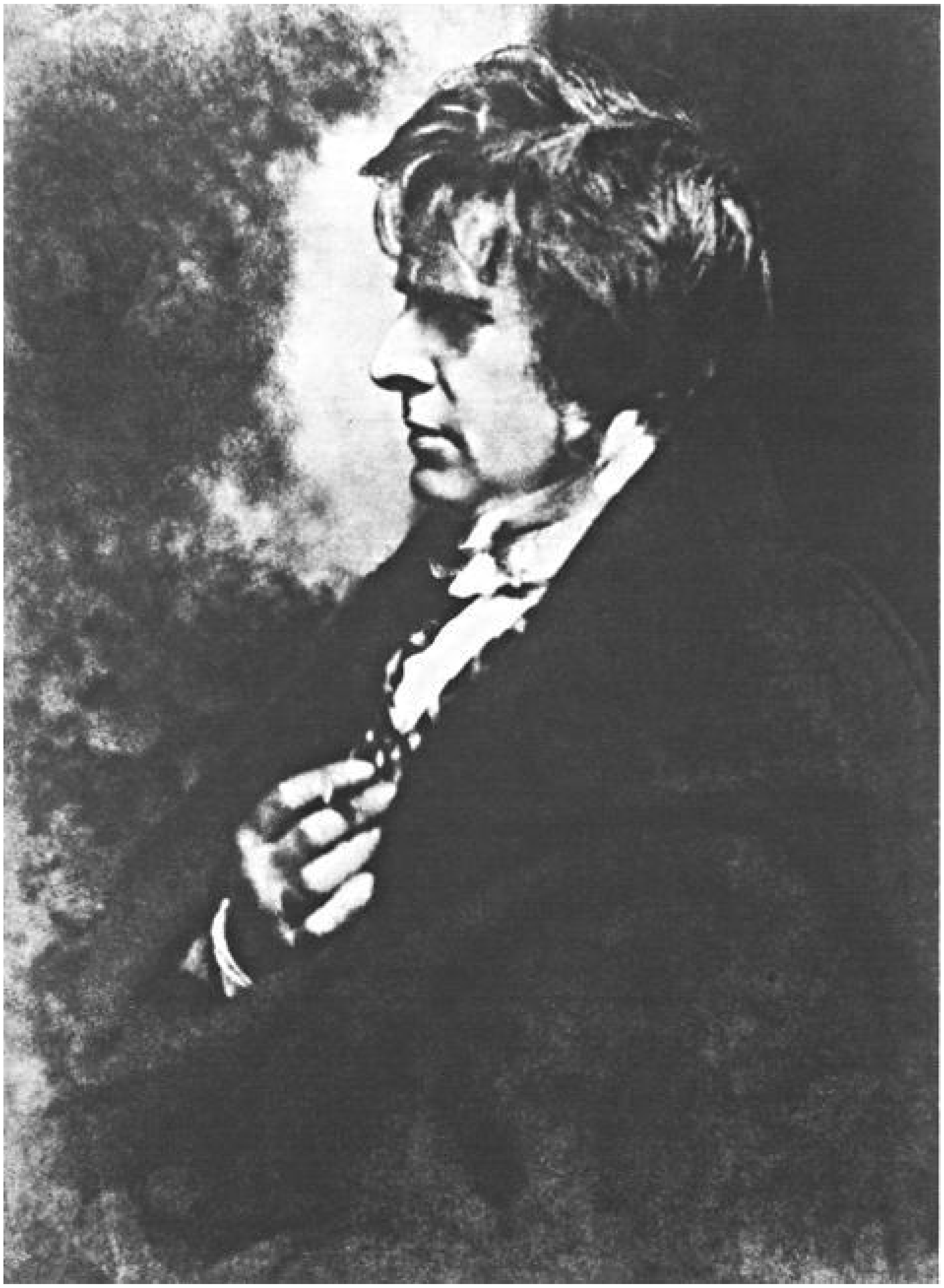
Louis Jacques Mande Daguerre (born 1789, died 1851), was a great French scene-painter who experimented for many years trying to find some way of rendering permanent the image projected by a lens. J. Nicéphore Niépce was engaged in the same research, and from 1829 until the death of Niépce in 1833 the two worked together, but it was not until some years after the latter date that Daguerre discovered the process that bears his name. This process may be briefly described as follows: a highly polished and perfectly clean silver plate is rendered sensitive to light by the formation of a deposit of silver iodide on the surface, this being accomplished by exposing the plate —of course in the dark—for some minutes to the vapor of iodine. When the plate has assumed a uniform golden-brown color it is placed in the camera and the exposure is made, the light projected by the lens causing a chemical change to take place in the silver iodide. The image thus obtained is very weak, and in order to strengthen it the plate is exposed for some minutes to the vapor of mercury. It is subsequently fixed, or rendered permanent, by bathing with a solution of sodium thiosulphate (ordinarily known to photographers as "hypo"). This dissolves the silver compounds that were not affected by light. In some cases the picture is still further strengthened by treating it with chloride of gold. This not only increases the vigor of the image but at the same time improves its stability, so that it is less likely to fade as the result of atmospheric action or exposure to light. The effect of the chloride of gold is literally to gold-plate the image. As the surface of the completed daguerreotype is very sensitive to any mechanical action, it must be protected by glass. A mere touch of the finger leaves an irremediable scratch.

The daguerreotype was at one time very popular for portraiture, but the process has certain drawbacks that have caused it to be superseded by improved methods. Among these drawbacks are the following: The exposures required are rather long; it is impossible to make duplicates—a separate exposure must be made for each picture; the picture must be held at a certain angle to make it visible, and the process is rather expensive and laborious to work. Nevertheless, exquisite effects may be obtained in daguerreotype; the writer has seen pictures of this kind which for sheer beauty yield to none of the modern printing mediums.

The decadence of the daguerreotype is to be regretted for at least one reason,—the man who elected to work in that medium was necessarily at least a craftsman, whereas at the present time many photographers are neither artists nor craftsmen, but merely mechanics of only fair skill. Photography has been brought to such a state of perfection that good technical results may be obtained by persons that work by rote and know absolutely nothing of the principles underlying the craft. This lack of training and enthusiasm for the work must evidently be reflected in the results obtained.

There are few forms of portraiture art that equal in beauty choice early examples of daguerreotype photography. They have the exquisite delicacy, softness and individual charm of the best miniature portraits. Good old daguerreotypes are treasured possessions in the homes of many families—and rightly so, for they combine a fine quality of art with a gentle personal appeal.

WRITTEN FOR THE MENTOR BY PAUL L. ANDERSON
ILLUSTRATION FOR THE MENTOR. VOL. 6, No. 12, SERIAL No. 160



FROM A PRINT BY ALVIN LANGDON COBURN

SELF-PORTRAIT PHOTOGRAPH—BY D. O. HILL

PHOTOGRAPHY

Portraits by D. O. Hill

TWO

David Octavius Hill (born 1802, died 1870), was a Scotch painter who conceived the idea of producing a great historical picture representing the Disruption of the Church of Scotland. This work involved painting some four or five hundred portraits, and Hill, despairing of obtaining satisfactory sittings from so many persons, turned to the newly discovered art of photography to furnish the portraits he needed, with the idea of using the photographs as a guide in painting. Hill used the calotype process, invented by Fox Talbot, which rendered a piece of paper sensitive to light by coating it with iodide of silver. When it was exposed in the camera and developed, a negative resulted, and positive prints were made from this negative in the same medium.

Hill became so much interested in photography that he worked with it for several years, to the neglect of his painting. During those years he produced photographic portraits which have certainly never been surpassed, and which some people think have never been equalled. The exposures necessary were very long—four or five minutes in bright sunlight. This fact lends a great deal of beauty to the results, for there is no doubt that full sunlight gives effects that cannot be obtained in any other way, and these may be of surpassing beauty, provided the photographer is skilful enough to manage his apparatus and pose the sitter properly. It is regrettable that so many photographers of the present day shun out-door portraiture, for there is unquestionably a great opportunity in that class of work. The claim of some photographers that out-door light is not satisfactory for portraiture is refuted by Hill’s results.

Hill was not a great painter. His works in that medium are well-nigh forgotten, but he was unquestionably a man of great sensitiveness, who possessed the quality of psychic insight so necessary to a portrait worker. It is the estimate of an authority that, though he could never be compared with the great masters of portraiture, Rembrandt and Velasquez, nevertheless his works are entitled to a place in the second rank.

Hill was especially fortunate in his sitters, for the men and women that he photographed were persons whom it would be difficult to render commonplace in appearance, among them being Christopher North (Professor John Wilson), J. G. Lockhart, Lady Ruthven, Robert Haldane, William Henning, Mrs. Anna Brownell Jameson, and others of equal note in Great Britain.

The paper negatives made by Hill are carefully preserved. The writer is fortunate in the possession of prints from two of these negatives. The reproduction shown herewith, a gum-platinum plate made and given to him by Alvin Langdon Coburn, is from one of them. Much of the beauty of this example of Hill’s work is due to modern printing methods, but the quality in this negative, brought out in the print, proves undeniably that Hill merits recognition as a master of portraiture.

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ILLUSTRATION FOR THE MENTOR. VOL. 6, No. 12, SERIAL No. 160

PHOTOGRAPHED AT YERKES OBSERVATORY, WILLIAMS BAY, WISCONSIN. OCTOBER 19, 1911

BROOKS' COMET—ASTRONOMICAL PHOTOGRAPH

PHOTOGRAPHY

Astronomical Photography

THREE

Photography has made possible many discoveries of tremendous importance in the realm of astronomy by revealing the existence of stars too faint—because of their small size or great distance—to be seen by the eye. This is one of the most conspicuous ways in which the sensitive plate has been an aid to the scientist. A device for carrying a photographic plate is attached to a telescope and the plate exposed to the image projected by the telescope for a prolonged period. This may, in fact, amount to several hours; exposures are sometimes partly completed one night and finished the next, a comparatively small area of the heavens being chosen for investigation at one time. On development of the plate the stars are counted and compared with existing charts of the area in question. Of course this method requires that the telescope move with the same angular velocity as that of the earth's rotation, so that the image of each star may remain in precisely the same position on the plate during the entire time of exposure. Otherwise the star would be represented as a trail of light, the slightest variation in the speed of rotation being sufficient to cause blurring of the image. It is apparent that the clockwork employed for driving the telescope must be a marvel of accuracy.

The power which this method possesses of revealing hitherto undiscovered stars depends on a curious fact. If an observer looks into the eye-piece of a telescope he can discern only those heavenly bodies that send to the earth a certain minimum of light; but when a photographic plate is exposed for long periods there is a cumulative effect of light on the sensitive emulsion. That is, the long-continued impact of the light rays causes, little by little, a gradual change in the constitution of the sensitive silver salt. The action thus piles up, so to speak, and records light that is far below the visible minimum.

The photographic plate has not only aided discoveries in the vast realms of interstellar space, but has also revealed to us things so exceedingly minute that no other method of observation could bring them within the range of our perceptions.

WRITTEN FOR THE MENTOR BY PAUL L. ANDERSON
ILLUSTRATION FOR THE MENTOR. VOL. 6, No. 12, SERIAL No. 160

BY PAUL L. ANDERSON

PORTRAIT—BY THE BROMOIL PROCESS

PHOTOGRAPHY

A Bromoil Print

FOUR

In the bromoil process, the first step is to make a bromide enlargement. The negative from which a print is to be made is placed in an apparatus resembling the familiar stereopticon and an enlarged image is projected on a piece of bromide paper, or paper that has been coated with an emulsion similar to that used for plates. After the paper has been exposed to the image it is developed, fixed and washed, the result being a large positive print of the original small negative. Often this print is allowed to remain as it is, and it is then known as a bromide enlargement, or, simply, an enlargement; sometimes the worker converts it into a bromoil.

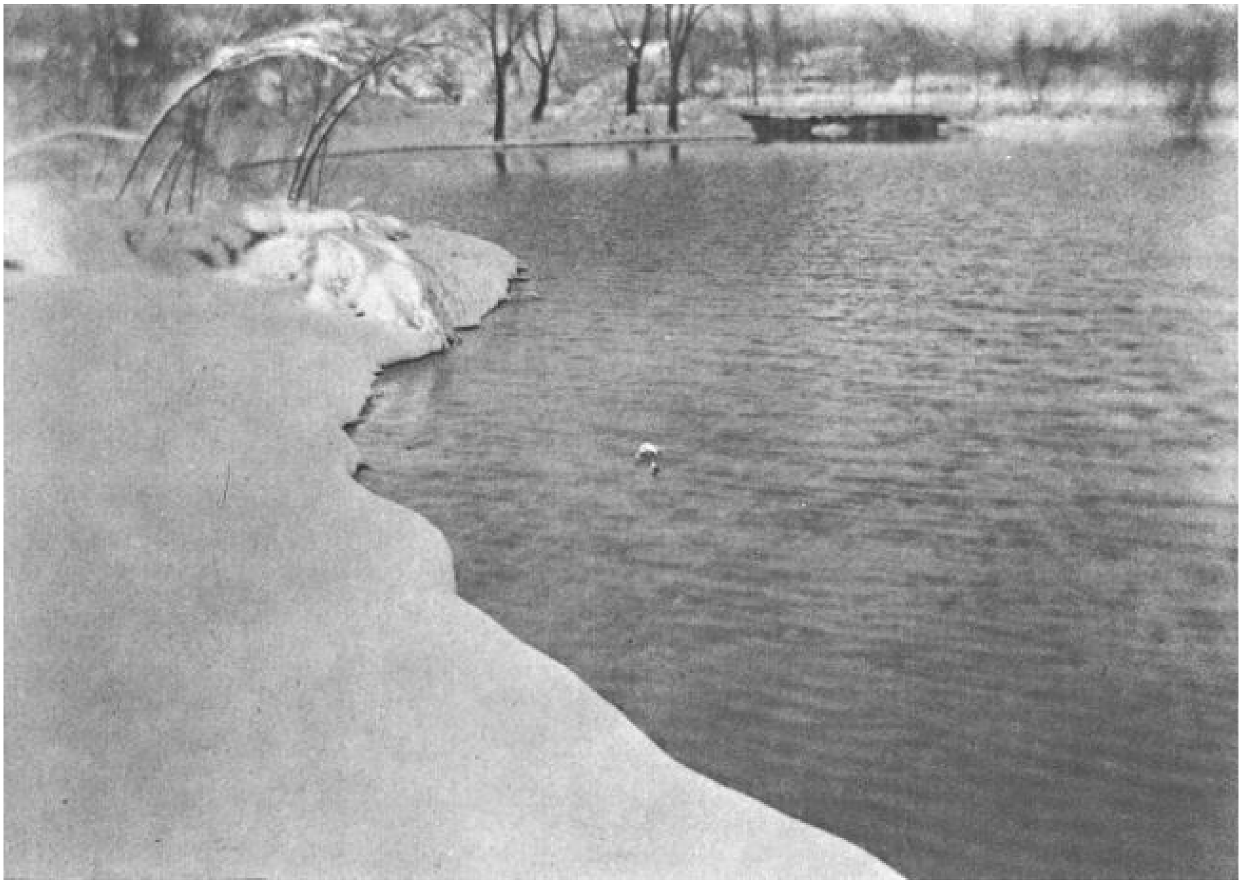
The image in an enlargement consists of metallic silver in a film of gelatine, the gradations of the picture resulting from the varying thicknesses of the silver image. The first step toward changing this to a bromoil is to treat it with certain chemicals that bleach out the silver image and at the same time harden the gelatine in proportion to the amount of silver present. The bleached print is then soaked in warm water, and the high-lights of the picture, where the gelatine is least hardened, absorb the water freely, the half-tones less so, and the shadows least of all. An oily ink, then dabbed on the print with a brush, adheres freely in the shadows, less freely in the half-tones, and least of all in the lights, being repelled by the water in the film. The final result is a print in which the image is formed by varying thickness of ink, which, of course, may be of any color.

The advantages of bromoil over bromide are numerous. In the first place, a bromide print cannot be regarded as absolutely permanent, but a bromoil may be. Next, the color of a bromide print is limited to black and varying shades of brown, unless chemical toning is resorted to, which still further reduces the stability of the image. But a bromoil may be of almost any color, and, indeed, of different colors in different portions of the picture. The greatest advantage of the bromoil process, however, lies in the fact that as much or as little ink as may be desired can be put on any given area. By varying the consistency of the ink it can be made to adhere more or less freely. By modifying the brush action it can be placed on the print or omitted from it, and can even, at times, be removed after being deposited on the paper. It will be seen that the artist has complete control over the gradations, and to some extent, also, over the outlines of the picture. He can therefore make the process respond to his desire for artistic expression to an extent not possible with any other photographic printing medium, even the superficial texture of the image being largely under the worker’s control.

A variant of bromoil is the oil process, though it would be more correct to put it the other way about, the latter process being the older of the two. A sheet of paper is coated with gelatine alone, this being rendered sensitive to light by means of certain chemicals and then printed under a negative. The effect is to render the gelatine hard in proportion to the amount of light-action, that is, hardest in the shadows, less so in the half-tones, and least of all in the lights. The print is then washed to remove the excess of sensitizer, and soaked in warm water; the subsequent operations are the same as in bromoil. Oil is superior to bromoil in being slightly easier to manipulate and in not requiring a dark-room, but it is inferior in that it demands either daylight or a powerful artificial light for printing. Furthermore, a negative the size of the finished print is necessary, whereas with bromoil, large prints can be made from small negatives.

Oil and bromoil have the drawback of not being very rapid to work, three or four 11×14 bromoils representing a good day’s work for a careful manipulator, but they are by far the most satisfactory of all photographic printing mediums when the desire is for artistic expression.

WRITTEN FOR THE MENTOR BY PAUL L. ANDERSON
ILLUSTRATION FOR THE MENTOR. VOL. 6, No. 12, SERIAL No. 160



PHOTOGRAPH BY PAUL L. ANDERSON

THE LAKE, WINTER—NATURE PHOTOGRAPHY

PHOTOGRAPHY

Pictorial Photography

FIVE

The accompanying photograph entitled “The Lake, Winter,” illustrates admirably the use of the soft-focus lens. It is also of interest as showing the advantages sometimes to be gained from the intentional use of defects. The normal human eye is unsurpassed for the purpose for which it is designed; it is difficult to imagine an organ more perfect in this respect. The eye automatically, and almost instantaneously, adjusts itself for near or distant objects and for varying intensities of light, and has, moreover, a field of view of nearly 180 degrees—almost a complete half-circle. Nevertheless, it has two defects that tend to impair the accuracy of vision, namely, chromatic and spherical aberration. Chromatic aberration is the inability to focus simultaneously on two or more of the primary colors (it is this defect in the eye that causes red letters to seem to stand out from a blue or green background, a trick sometimes used in poster work). Spherical aberration is the inability to bring to a focus the rays of light that pass through the lens near the margins at the same time as those that pass through near the center. For these reasons—and, in lesser degree, some others—the eye cannot see sharp lines, and a lens that gives sharp definition to the edges of objects produces results that are esthetically unpleasing, because foreign to our experience. The soft-focus lens—of which there are numerous makes—is so designed that it possesses the errors that are normal to the eye, and therefore—if the characteristic softness of definition is not over-done by a too enthusiastic worker—gives results having an agreeable vagueness of outline. At one time the qualities of this type of lens were over-worked, the results being so excessively diffused that, as one writer said of a print, “it was impossible to tell whether it was a ‘Portrait of a Lady’ or a ‘Water-Spout in the Gulf Stream.’” But for some years past the pendulum has been swinging the other way, and photographers in general (it must be understood that this refers only to artistic workers, not scientists) are now using the unconnected lens so as to secure as nearly as possible the quality characteristic of the normal eye with, perhaps, a slight exaggeration for the sake of suggestion, and as a stimulus to the imagination.

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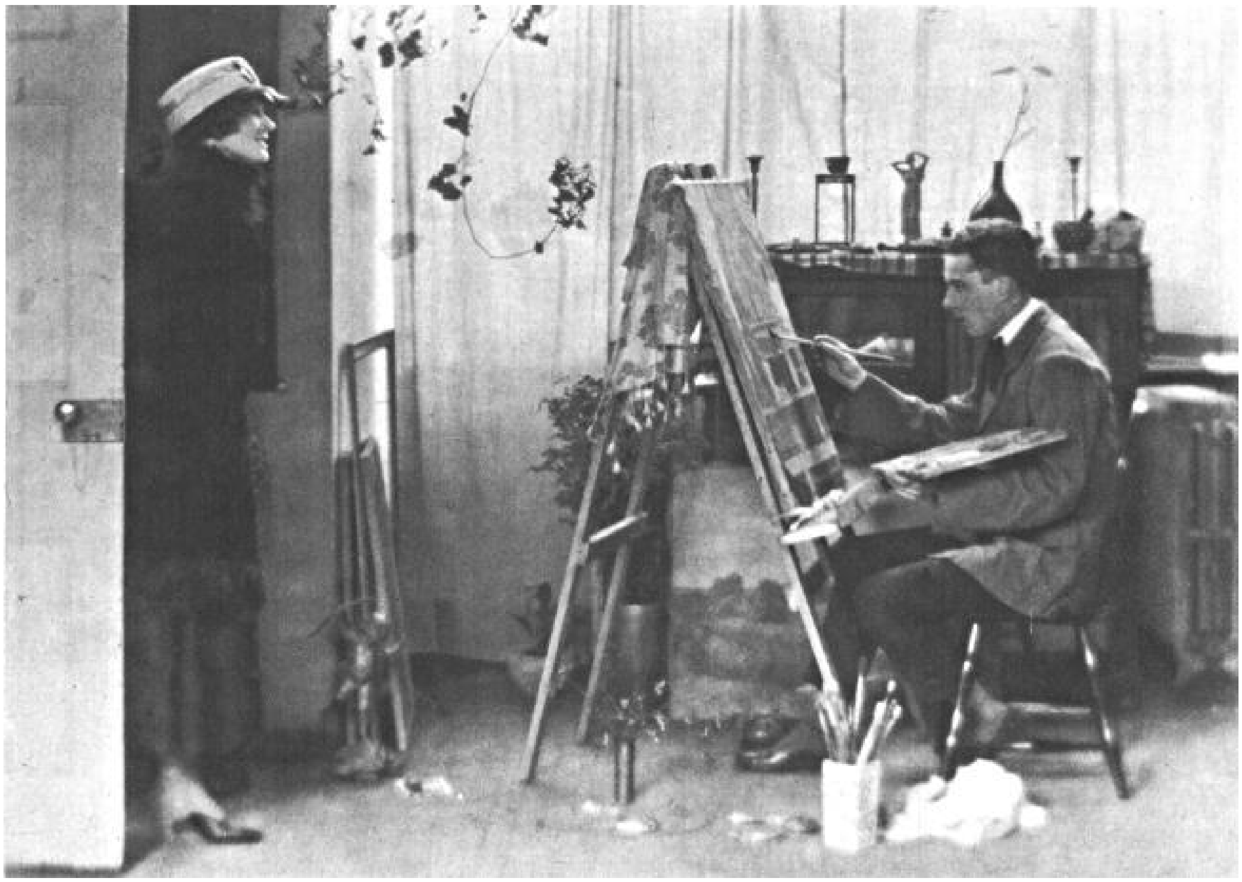
ILLUSTRATION FOR THE MENTOR. VOL. 6, No. 12, SERIAL No. 160



PHOTOGRAPH BY KARL STRUSS, BY COURTESY OF HARPER'S BAZAAR.

PHOTOGRAPH ILLUSTRATION FOR A STORY

PHOTOGRAPHY

Motion Picture Photography

SIX

A phase of photography that has a very broad scope is motion-picture work, the mechanics of which depend on this fact: If an object is looked at for a time and is suddenly removed from before the eye, the eye continues to see it for an appreciable time after its removal. This phenomenon is called "persistence of vision." A motion-picture camera is so arranged that a long strip of film can be drawn past the lens in a series of jerks, the shutter being opened to permit the image projected by the lens to fall on the film during the period that the latter is at rest; the film is drawn on to the next position while the shutter is closed. Naturally, an object moving before the lens will move slightly during the interval between exposures, so the film, when developed, shows a consecutive series of photographs of the object in slightly different positions. A positive print is made from this series of negatives on a similar strip of film. This is projected by means of an apparatus something like the familiar magic lantern or stereopticon, but so arranged that this film may be drawn along in jerks. Each photograph is shown for a fraction of a second, and is replaced, during the time that the shutter is closed, by one showing a slightly later phase of the motion. Because of the persistence of vision the eye blends these successive photographs into one apparently continuous motion. It will be seen that the term "moving pictures" is really a misnomer, since the pictures on the screen do not move, but remain perfectly stationary during the time that they are seen. By taking the pictures rapidly and projecting them slowly the apparent motion may be slowed down, so that a rifle bullet may take three or four minutes to travel across a screen space of as many feet. By taking them at wide intervals and projecting rapidly the motion may be speeded up, and a plant may seem to grow from a seedling to maturity in a few minutes. The ordinary taking and projecting speed is sixteen pictures per second, experiment having shown that this is the least number that the eye will blend satisfactorily. Since each picture is one inch wide by three-quarters of an inch high—in the film—it is evident that each second of time represents one foot of film. The writer has seen a rather elaborately staged photo-play that required an hour and forty minutes for projection; a simple calculation shows that this involved 6,000 feet of positive film—a little over a mile. The length of the negative film was undoubtedly more, on account of retakes, cuts, and so on. An expenditure of five or six hundred dollars for film, however, is but a small item in the cost of producing an elaborate photo-play, for the actors receive large salaries—though not so large as the press-agents would have us believe—and the cost of scenery is great. The production of photo-plays is nevertheless a profitable industry, as may be understood from the fact that the average daily attendance in this country is estimated at about twelve million. Assuming that each spectator pays only ten cents, this represents an intake of $1,200,000 daily and, as is well known, the prices of admission in many theatres range from 25 cents up to $1.00 and more. The artistic possibilities of the motion-picture play are obviously limited—it can never hope to rise to the emotional heights of the legitimate drama—but they are none the less considerable. It is to be regretted that the motion picture industry is at present so much in the hands of producers who pander to the coarser instincts of the public, through sensationalism and slap-stick farce; who are often indifferent to detail—the writer has seen a cow-puncher represented as wearing his six-shooter butt-foremost; who treat the author’s work according to their own ideas. A well-known author remarked, on seeing the screen version of one of her books: “If I hadn’t been fairly familiar with the story I wouldn’t have known what it was all about.” In general, firms seem to be more concerned with getting the public’s money than with producing really artistic results. The writer once saw a photo-play version of a fairly well-known book, in which the producer had changed an elderly, gray-haired, quiet, experienced cattleman into a cheap imitation of a Bret Harte gambler of thirty years of age, the purpose of this metamorphosis being to transform a noble and self-sacrificing affection into a piece of gaudy sensationalism. Such tactics cannot fail to displease thinking people, but there are, fortunately, producers to whom these remarks do not apply—really conscientious men of high ideals, and signs are not wanting of an improvement in this regard. The motion picture, in worthy hands, can be made an educational medium of great value, not only in the dramatic art but in many other ways. Films frequently show scenes of historical interest, life in foreign lands, industries. Scientific subjects are treated, such as the peristaltic movements of the intestines and the action of the heart, photographed by means of the X-ray; also the life cycle of micro-organisms, the microscope being used in this case—and many other activities of life. Among the most interesting of these films are those produced by the Williamson brothers, showing sea life, though mechanical difficulties have so far prevented the photographing of the most interesting phase of marine life, that of the extreme depths.

WRITTEN FOR THE MENTOR BY PAUL L. ANDERSON

ILLUSTRATION FOR THE MENTOR. VOL. 6, No. 12, SERIAL No. 160

THE MENTOR ·· AUGUST 1, 1918
DEPARTMENT OF FINE ARTS

THE STORY OF PHOTOGRAPHY

By PAUL L. ANDERSON

Artist Photographer, Author of “Pictorial Photography”

MENTOR GRAVURES

PHOTOGRAPH FROM A DAGUERREOTYPE

SELF-PORTRAIT PHOTOGRAPH *BY D. O. HILL*

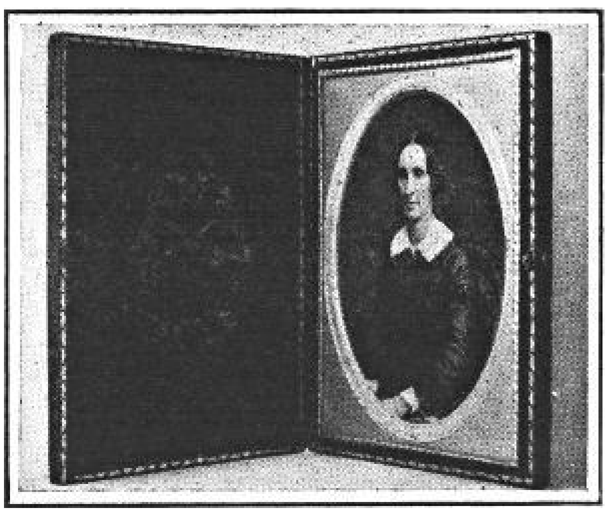
BROOKS’ COMET

MENTOR GRAVURES

PORTRAIT BY BROMOIL PROCESS

THE LAKE, WINTER

ILLUSTRATION FOR A STORY



A DAGUERREOTYPE

Entered as second-class matter March 10, 1913, at the postoffice at New York, N. Y., under the act of March 3, 1879. Copyright, 1918, by The Mentor Association, Inc.

Numerous investigators, Daguerre, Niépce, Fox Talbot, and others, have been credited with the discovery of photography, but the fact seems to be that these, and many more, merely contributed, each in his turn, some portion of the total that goes to make up the art as it now stands. Photography means, literally, “light-writing,” the name being derived from two Greek words, *phos*, light, and *graphein*, to write. The practice of photography depends primarily on the fact that certain chemical compounds are changed into other compounds by the action of light. Another fact is closely allied with this, namely, that a suitably constructed lens of glass or other transparent material, or a fine needle-hole used instead of a lens, will project the image of objects placed in front of it. A camera, then, consists of a light-tight box having at one end an arrangement for holding a lens or a card with a needle-hole in it, and at the other end a device for holding some light-sensitive chemical to receive the image projected by the lens. In modern practice this light-sensitive chemical is almost always bromide of silver or a mixture of the bromide with other silver compounds, these chemicals being held in an emulsion of gelatine. When the gelatine emulsion is coated in a thin film on a sheet of glass the result is known as a dry-plate, or, simply, a plate. When it is coated on a strip of celluloid wound on rollers so that successive portions may be exposed to light, it is called a roll film, and when it is coated on separate sheets of celluloid, arranged like a pad, to be exposed successively, it is called a film pack. A similar emulsion coated on paper gives bromide or gas-light paper, which, as will be seen later, is used for making prints. At one time wet collodion plates were generally used, a sheet of glass being coated with collodion and sensitized by bathing it with iodide of silver. The exposure was made before the plate dried; but these plates were inconvenient to handle and have been almost entirely superseded by the gelatine dry plate. The prepared plate, of whatever type it may be, is placed in the camera and exposed for a longer or shorter time, depending on circumstances, to the light projected by the lens, but no image is visible after exposure, (unless, indeed, the exposure has been tremendously excessive,) and the plate must be developed.

There are about fifty different reducing agents on the market; most of them are derived from coal-tar, though some are made from nut-galls, lichens, and other substances. The developer consists of a solution in water of one or more of these reducing agents, with other chemicals to control the action, the exposed plate being bathed in this solution, either in the dark or in a light to which the plate is not sensitive. Wherever light has acted on the silver salts the developer causes metallic silver to be deposited in proportion to the amount of light-action, so that on holding the developed plate up to the light a dense deposit is seen in those parts representing the brightest portions of the subject, while the shadows of the original are represented by

From a platinum print by Gertrude Käsebier

“BLESSED ART THOU AMONG WOMEN”

Portrait Photograph

From a photograph by Paul L. Anderson

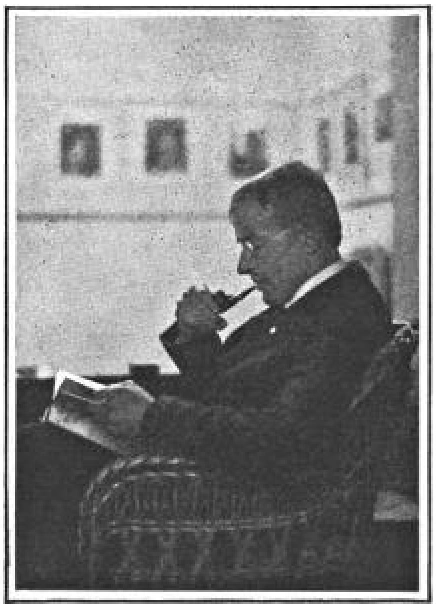
A COUNTRY STATION MASTER

Portrait Photograph

thin areas, and the half-tones by deposits of intermediate density. For this reason the developed plate is called a negative. The plate is then bathed in a solution of sodium thiosulphate (generally called “hypo”), which dissolves the unaffected silver salt but does not affect the metallic silver—or at least does so only very slowly. Next, the plate is washed in water to remove all unnecessary chemicals, and is dried. The ordinary plate is sensitive only to ultra-violet (invisible) and violet light, so it cannot render truthfully any subject having color, but by the addition of certain aniline dyes to the emulsion it may be rendered sensitive to green in addition to violet and ultra-violet; it is then described as orthochromatic (“right-colored”) or isochromatic (“equally-colored”). Still other dyes extend the sensitiveness to include not only ultra-violet but also the entire visible spectrum; such a plate is called pan-chromatic (“all-colored”).

Printing the Photograph

The finished negative, when dry, must of course be printed, and there are many printing mediums available. The carbon process gives an image in lamp-black or some earth pigment, bound up in a film of gelatine; the gum-pigment process gives an image similar to that of carbon, the binder in this case being gum arabic; the platinum process gives an image of black



From a photograph by Paul L. Anderson

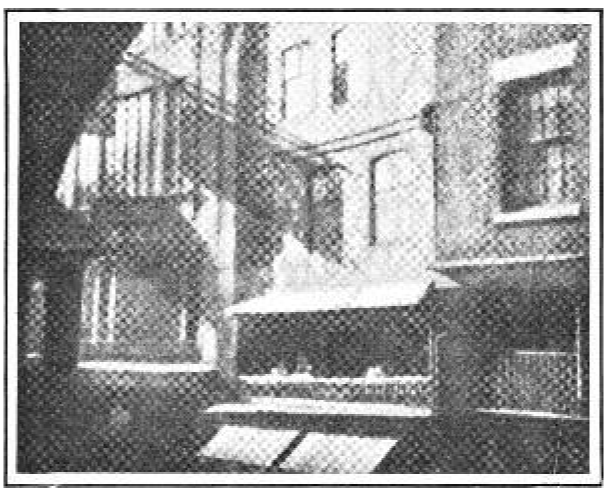
A PORTRAIT PHOTOGRAPH

metallic platinum direct on the paper support. Other processes give different effects, one of the most valuable to the pictorial worker being gum-platinum, in which a completed platinum print is coated with a gum-pigment film and printed under the negative a second time, the final result being a gum-pigment image superposed on the platinum image. Of all printing mediums the one that has most intrinsic beauty, and is at the same time most capable of rendering satisfactorily the gradations of the negative, is probably platinum, so this is most used by pictorial workers. But, since it is expensive and requires daylight or strong artificial light for printing, nearly all commercial workers prefer the somewhat less beautiful and less permanent, but more convenient, gas-light paper, so-called because it can be manipulated entirely by gas-light, neither daylight nor a dark-room being required. This medium consists of paper that has been coated with an emulsion somewhat similar to that used for plates, but requiring much longer exposure. The negative is placed in an appliance that holds it in close contact with the paper, then a sheet of paper is put in, and an exposure of a few seconds is given. Obviously, the paper receives most light under the thin parts of the negative and less under the denser portions, so that when the print is developed, fixed, washed and dried the resulting picture is light where the original subject was light, dark where that was dark, and show intermediate gradations where these existed in the original.

For purposes of reproduction two processes depending on photography have almost entirely superseded the older methods of etching and wood engraving. These photo-mechanical processes, as they are called, are far more rapid and much cheaper, and are, in addition, more accurate. In photo-gravure the photographic image—copied by photographing the original—is transferred to a copper plate and the plate is automatically etched in an acid bath to varying depths, depending on the depth of shadow in the original. This plate is then inked all over, the ink being worked into the depressions

Photograph by Karl Struss

CAPRI, ITALY



From a bromoil transfer by Charles Kendall

THE SHADOW OF A SIGN

in the copper, and the surface ink wiped off. A sheet of paper is brought into contact with the plate under heavy pressure, and, being forced into the hollows of the copper and taking up the ink from them, a print results. In the less beautiful but cheaper and more rapid half-tone process the copy is made through a cross-ruled glass screen, the image being thus broken up into a series of dots. The image so obtained is transferred to a zinc plate, which is etched in an acid bath or with an acid spray. The dots serve to protect the zinc from the action of the acid.[A] The finished plate shows an image consisting of dots with hollows between them, the dots being large and near together in the shadows, small and far apart in the lights. This plate is inked with a roller, and a sheet of paper, lightly pressed against it, takes up the ink to form a print. Thus it will be seen that photogravure is an intaglio (cut-in) process, and half-tone a surface-printing process.

[A] See cut on page 9.

Photographic Illustrations

Photography has not only superseded manual processes for reproduction, but has also largely replaced them for purposes of illustration. Practically all news illustrations are now made by photography, which is also extensively used for advertising work. To a less extent it is employed for fiction illustration, admirable work having been done in this field by Clarence H. White, Karl Struss, and Lejaren à Hiller. It does not, however, seem probable that photography will ever entirely replace draughtsmanship for the illustration of fiction, since the very strength of the camera,—that is, its surpassing power of rendering accurately the outlines and gradations of natural objects,—operates as a severe limitation in the case of original, imaginative work. It is difficult to conceive of "The Fall of the House of Usher" or "The Rime of the Ancient Mariner" being satisfactorily illustrated by photography, and if, for instance, "Le Morte d'Arthur" were made a photographic subject, the cost of models, costumes and scenery would probably be excessive.

From a platinum print by W. E. Macnaughton

SCENE ON THE CONNECTICUT RIVER

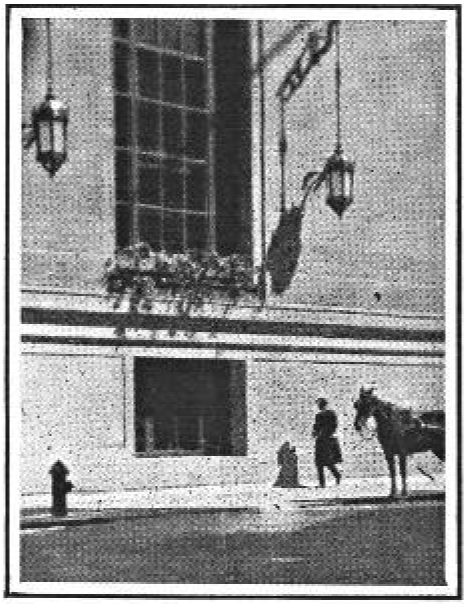
Despite the limitations of the camera as regards imaginative work there is a small but devoted band of photographers who use the camera as a means of artistic expression, and these men and women have produced some wonderfully fine results that fulfil the definition of art: "A means of arousing an emotion in the spectator." In the last analysis, however, it will be found that such results are due to one of two methods of approach: either the careful selection of an unusual natural effect, or the use of one of the so-called "control processes"—that is, printing mediums that allow the worker to modify at will either the outlines or the gradations of the negative, or both. In the former case,

From a carbon print by H. Y. Sümmons

"THE STYGIAN SHORE"

however, the photographer cannot be regarded as more than an exceptionally sensitive and perceptive craftsman, and in the latter instance the camera user, of course, ceases to be a photographer and becomes a creative artist, using photography merely as a basis on which to construct an imaginative result. The possibilities of this second method of work have not yet been fully explored; they appear to be limitless.

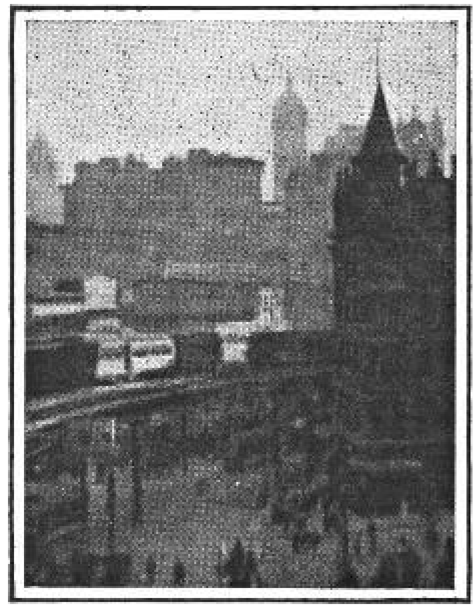
The Precision of Photography



Photograph by Struss

AFTERNOON SUN

The Biltmore, New York



Photograph by Struss

A MISTY DAY

Chatham Square, New York

Photograph by Struss

SUNSHINE AND SHADOW

Park Row, New York

The literalness of photography, which prevents its ever competing with etching or painting in imaginative art, makes it of inestimable value in

Photograph by Struss

SUNLIGHT ON SURF

certain realms, and scientists of all sorts, astronomers, physicists, physicians, pathologists, as well as architects, building contractors, business men, who wish a precise and accurate record of any object, recognize the value of the camera. Photographs are often admitted as legal evidence in court. It is impossible to overstate the value of the dry-plate to the surgeon, since the X-ray, generated by passing an electric discharge through a glass tube from which most of the air has been exhausted, penetrates many objects that are opaque to ordinary light, and, though invisible to the eye, nevertheless affect a photographic plate, thus making possible a precise diagnosis of fractured bones, gun-shot wounds, digestive disturbances, and many other pathological conditions in which diagnosis without a radiograph would be mere guesswork.

In portraiture, photography is superior to any other graphic art, since the camera worker can, by judicious selection of lighting, pose and facial expression, render the character of the sitter quite as well as the draughtsman, this being the final test in portrait work, though it must be admitted that few portrait photographers meet this requirement.

Microphotography

The human eye and mind are, from a mechanical point of view, but imperfect instruments. Admirably as they serve the purpose for which they are designed, it is nevertheless impossible for them to observe with absolute accuracy. The camera, however, has no such limitations; its observations are accurate and its records unquestionable, so long as no definite effort is made to impair their exactness. For this reason photography is used not only in astronomy but in many other branches of science, among its most important uses being the making of records of microscopic objects.

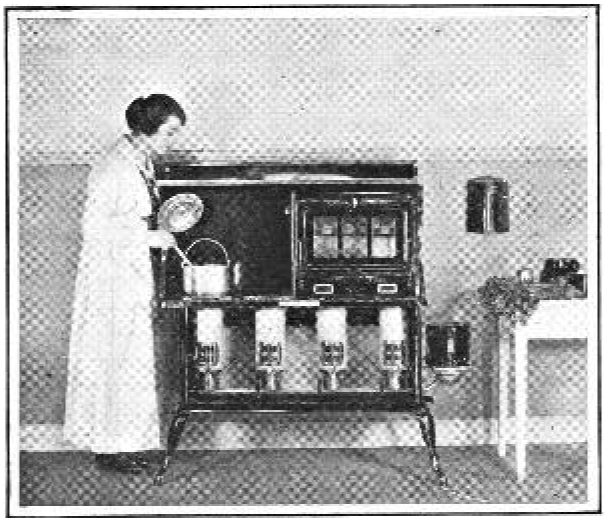
A device carrying a photographic plate is attached to the eye-piece of a microscope; the plate being exposed affords, on development, a precise record of the subject under observation. It may be noted that in this case, as in astronomical photography, no camera lens is required; the microscope, like the telescope, projects an aërial image which is impressed on the plate. It thus becomes possible for the microscopist to study at leisure a photograph of the object that was in the field of the microscope, and thereby eliminate eye-strain and minimize the likelihood of overlooking any feature of interest. It is further possible to make lantern-slides from the negative so obtained. A lecturer by this means is enabled to show the photograph to a large group of individuals simultaneously.

By Paul L. Anderson

PHOTOGRAPH TAKEN IN A SNOWSTORM

New York

Though the photographic plate thus extends the usefulness of the microscope, this is not the limit of its value in this respect. Light is transmitted by waves, similar in some ways to waves in water, the light waves being disturbances of the light-bearing ether, an invisible, imponderable substance of zero density and infinite elasticity, which pervades all matter. (It must be understood that the ether has never been observed nor its actual existence proven; it is, however, a necessary assumption for the satisfactory explanation of the observed phenomena of light, so far as our present knowledge extends.) The distance from the crest of one wave to the crest of the next is known as the wave length, the lengths of the various light waves having actually been measured. The human eye is sensitive only to waves between about four-ten-thousandths and seven-ten-thousandths of a millimeter in length, a millimeter being about one-twenty-fifth of an inch, and an object is invisible in the microscope if its diameter is less than half the wave length of the light by which it is illuminated, since in that case the light waves bend around the object and meet on the other side. We cannot, therefore, see objects whose diameter is less than about two-ten-thousandths of a millimeter. But the photographic plate is sensitive



Photograph by Struss

By courtesy of Cleveland Metal Products Co.

ADVERTISING ILLUSTRATION

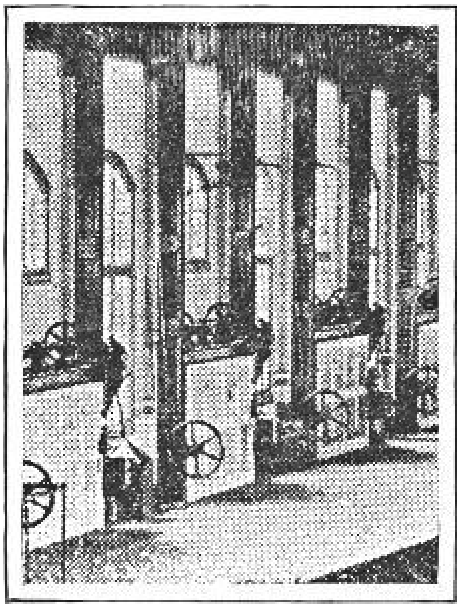
An example of commercial photography

THUMB-PRINT ON DARK WOOD

Finger-prints are made visible by dusting with a fine powder, and are photographed with a special detective camera

to shorter waves than the eye; these waves are known as the ultra-violet. By illuminating the microscope stage with ultra-violet light it therefore becomes possible to photograph objects so small that they must forever remain invisible to the naked eye, unless, indeed, the progress of human evolution brings with it increased sensitiveness to the shorter wave lengths. In this connection it is interesting to note that there are organisms so small that they cannot be made apparent to us even by photography, though we are made aware of their existence by inductive reasoning from their observed effects.

In the case of some objects, a fuller knowledge of their character is gained if they are examined in a manner somewhat different from that usually adopted. One of the photographs given herewith shows the effect obtained by what is known as “dark ground illumination.” Ordinarily, the light by which a microscopic object is examined passes through the slide, so that an opaque object is really seen only as a silhouette, but in dark ground illumination an opaque background is placed behind the object, and the light is allowed to fall on it from the sides. The object is thus made visible by the light that is refracted (that is, bent) into the lens of the microscope. In



NEWSPAPER ILLUSTRATION

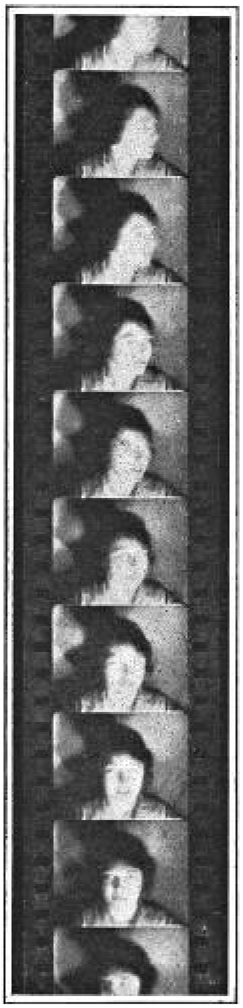
Enlarged to show half-tone screen

the present instance the effect seen by looking into the eye-piece was wonderfully beautiful, the crystals glowing with a brilliant yellow light against an intensely black ground.

Radiography

Some persons object to the inclusion of radiography as a branch of photography, since no camera or lens is used, but "photography" means, literally, "light-writing," and radiography is precisely this.

If the air be nearly exhausted from a glass tube, so that a high vacuum exists therein, and it be then sealed up, a current of electricity may be sent through the remaining air, setting up ether vibrations that pass out from the tube. These ether waves have the power of passing through many substances that are opaque to visible light, the X-rays, as they are termed, being totally invisible, though light waves to which the eye is sensitive are set up at the same time within the tube. Many persons confuse the greenish light from an X-ray tube with the X-rays, but the two are actually entirely different manifestations. The X-rays, though invisible to the eye, are nevertheless able to affect a photographic plate strongly, so that photographs may be made through solid objects. For example, if a sensitive plate be laid on a table and the arm or the hand placed on it, and an X-ray tube is brought near the arm, a photograph results in which the bone is represented as a dark area and the flesh around it as lighter, this being, of course, simply a shadow picture. This affords an intensely valuable aid to diagnosis, and a good surgeon will, if possible, first radiograph a fractured bone before setting it, unless the circumstances are very exceptional. The value of radiography is not, however, confined to fractures, but extends to wounds (it is of great help in locating metallic fragments or other foreign bodies in a wound), to many intestinal disorders, and to the diagnosis of other diseased conditions.



STRIP OF MOTION-PICTURE FILM

Actual size. The photography consumed five-eighths of a second, or just long enough for the subject to turn her head

Though not strictly bearing on photography, it is interesting to note that the X-rays, like the "gamma rays" (γ-rays) of radium, are in reality ether vibrations of very short wave length, and like the shorter waves (the ultra-violet) in sunlight, possess curative powers in some skin disorders and also the power of causing terrible burns. Sunburn does not result from exposure either to visible sunlight or to the heat of the sun, but to the ultra-violet rays; and an X-ray burn is identical with sunburn. In extreme cases the X-rays may cause complete destruction of the skin and even cancer, and before the properties of the X-rays were so well known as at present many operators lost hands, and some their lives, as a result of excessive exposure to the rays. At present, X-ray workers shield themselves, and, when necessary, the patient, with lead screens, that metal being practically opaque to the rays.

(A)

By courtesy of W. Faitoute Munn

(B)

By courtesy of W. Faitoute Munn

MICROPHOTOGRAPH OF CRYSTALS OF BRUCINE SULPHATE

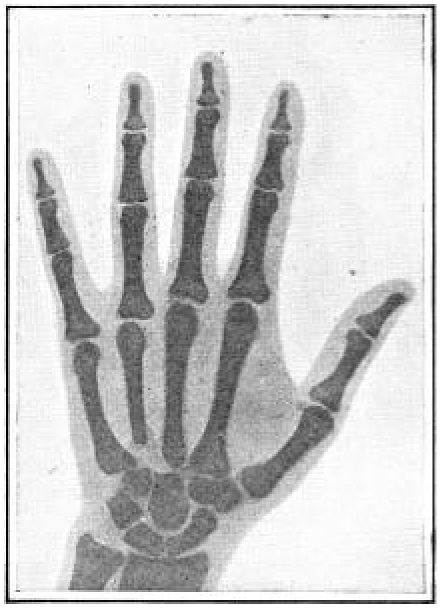
(A) By transmitted light

(B) With dark-ground illumination

Color Photography

Many workers have tried, with varying success to devise a means whereby photography could be made to reproduce not only the outlines and gradations of natural objects but the colors as well, and there is now available a method of great worth for this purpose. In brief, it consists in making, by one exposure in an ordinary camera, a set of three-color negatives, each of which represents that portion of one of the primary colors—violet, green and red—which was reflected from the subject. That is, one negative represents the violet “sensation,” the second the green, and the third the red. Prints are made from these negatives in suitable dyes on transparent films, which are cemented together, one over the other, thus giving a true color photograph, in which the secondary and tertiary colors—blue, yellow, orange, purple, brown, etc.—are obtained, as in painting, by the mixture in proper proportions of two or more of the primaries. This is the first method of color photography to possess the great advantage of producing prints—not transparencies, so that any number of duplicates may be made. No special camera is required, and the process is within the reach of any careful amateur. The writer believes the artistic value of color photography is relatively slight—a black and white art is capable of the fullest intellectual expression, and color is merely sensuous in its appeal. After much experiment with different color processes, he finds his own monochrome (single tone) prints more satisfying than the color work. However, the value of color to the scientific worker is incalculable, as will be realized at once on considering only one of the possible applications —namely, the study of skin affections. It is interesting to note that several methods have been devised for the reproduction of natural colors in motion-picture work—the familiar method of coloring the positive film by hand being only an approximation to truth. But none of those presented up to this time is fully satisfactory, though the prospects of future development are good.

When we consider the manifold and widespread uses of photography and the pleasurable diversion that it affords, it seems safe to say that there is no other form of industry not an actual necessity that is of such importance to the welfare and happiness of the human race.



By courtesy of Dr. T. W. Harvey

RADIOGRAPH OF NORMAL HAND

SUPPLEMENTARY READING

PICTORIAL PHOTOGRAPHY, PRINCIPLES AND PRACTICE	*By Paul L. Anderson*
INSTRUCTION IN PHOTOGRAPHY	*By Sir William de W. Abney*
SCIENCE AND PRACTICE OF PHOTOGRAPHY	*By John R. Roebuck*
PHOTOGRAPHY OF TODAY	*By H. C. Jones*
THE ARTISTIC SIDE OF PHOTOGRAPHY	*By A. J. Anderson*

⁂ Information concerning the above books may be had on application to the Editor of The Mentor.

THE OPEN LETTER

One summer afternoon, some years ago, I went into a front room of my home and drew down the window shades to shut out the glare and heat. The room became quite dark, but, in one of the shades, there was a small hole, through which the sunlight penetrated—casting on the white wall opposite, vivid images of all the objects in the street outside. I had before me a full-color, moving picture of the panorama of life that was passing the window.

Here was the original "camera obscura" ("dark chamber"). If one placed in the small hole in the shade a glass lens to give a sharp image, and substituted for the wall a movable screen, on which the projected objects could be focussed, one would have the essential elements of a modern camera. Through just such simple experiences as this important scientific discoveries are sometimes made.

* * *

For many years photography was largely confined to portraiture and the faithful reproduction of objects and scenes. All that was expected of a camera was to "make a picture" of a thing. Within the last forty years, however, as reproductive processes have been invented, photography has come to be one of the most useful of arts. Beginning about 1883, the quality and character of the illustrations in our magazines and books changed radically. Where, previously, there had been nothing but hand engravings of one sort or another, *photo-engraving* appeared, and, with that, the horizon of magazine illustration extended far beyond the reach of the liveliest imagination. Who could have foreseen, then, in the first photo-engraving processes, such possibilities as photographic printing in full colors, or moving picture films? Today, pictorial illustration depends on photography, and there is apparently little or nothing in life beyond the reach of photographic art. It discloses the internal arrangements of human anatomy;

it makes a record of the affairs of the heavenly bodies; it pictures things that the human eye cannot see; it is even potent in the realm of mystery, for have we not seen photographs of ghosts(?) reproduced from spirit seances? When objects and situations in life that do not exist are wanted, the camera can, by some trick or device, create them for us. There seems to be no limit to the possibilities. Each wonder displayed in photographic reproduction gives way to some new effect more wonderful still.

* * *

Consider briefly a few of the wonders of modern photography. First and foremost, and most spectacular of all, is the moving-picture film. Then, in the world of practical things, we have the telephoto-lens—a combination of the telescope and camera—that takes pictures of objects far beyond the reach of the naked eye. This enables one to photograph the head of a gargoyle on a distant cathedral, or the fledglings in an eagle's nest, or a mountain goat on a crag high up a mountain side. Then there is the swinging camera, by means of which a wide sweep of view can be included in one plate. A device of great practical value is photo-telegraphy, by which portraits for purposes of identification can be sent by cable and by wireless. In modern warfare the uses of the camera are many and varied. They include panoramic photography, photographing by moonlight, photographing of projectiles in the air, even photographing the noise of a gun by recording the vibrations due to the displacement of the air, photo map-making, photo surveying from the air, and the aviation gun camera. Radiography, too, must be mentioned—the X-ray and its use in surgery.

* * *

While all these wonders have come to pass in practical service, photography has likewise grown and expanded in the field of fine art. There are photographic art schools, and clubs and exhibitions—all for the purpose of cultivating and developing the camera to the finest forms of expression. We have highly cultivated and skilled photographers who are true artists, and who are engaged in employing photography as a means to fine art achievement. Among such artist photographers in this country mention should be made of Mr. Paul L. Anderson—the author of the present article;

Arnold Genthe, who, besides his wonderful portraits, has, by his art, preserved for future generations the scenes of old San Francisco—especially Chinatown—that have now passed away; Gertrude Käebier, Baron de Meyer and Jan de Strelecki, Stieglitz, Eyckmeyer, Steichen, Sümons and so many others that the list would fill this page.

W. D. Moffat
EDITOR

What, Who, and When?

WHAT IS PHOTOGRAPHY?

It is the science and art of producing pictures by the action of light on chemically prepared (sensitized) plates or films.

WHO DISCOVERED PHOTOGRAPHY?

No one particular individual. There is no known date on which “photographic action” was first recorded. The action of the sun in making impressions of one sort or another on surfaces was known to man from the earliest times. Records of it can be seen in fossilized vegetable remains—and this action of the sun is apparent in the change of color that takes place in the ripening of fruits and foliage.

WHO FIRST APPLIED A SENSITIZED PLATE TO THE PURPOSE OF MAKING PHOTOGRAPHIC PRINTS?

No single individual discovered this essential principle of photography. It came to be recognized in the course of many experiences, beginning with the alchemists; and developing through the experiments of a number of investigators, until the end of the eighteenth century, when the sensitiveness of various silver compounds to light became well known, and the character of the change produced on these compounds by light became established. Thomas Wedgwood, the fourth son of Josiah Wedgwood, the renowned potter, developed a process by means of which the image printed by photographic means could be “fixed” and made permanent.

WHAT INSTRUMENT BROUGHT THE PHOTOGRAPHIC PROCESS TO A PERFECTED FORM?

The camera. The camera is the photographic apparatus in which the image is projected upon the sensitized plate, thus securing a photographic impression. The word “camera” is Italian for “room,” and the full name of the original instrument, “camera obscura,” means “dark room.”

WHO INVENTED THE CAMERA?

Giovanni Baptista della Porta, who lived in the sixteenth century, has often been stated to have invented the camera, but he appears only to have popularized and improved it. The first use of cameras was not for printing photographs, but simply as an interesting toy or to assist one in tracing the outlines of various objects. There are many applications of the “camera obscura”—a notable one being the periscope of a submarine. It was not until a suitable sensitive plate was discovered that the camera became useful as an apparatus of photography.

WHO WERE MOST PROMINENTLY IDENTIFIED WITH THE DEVELOPMENT OF THE PHOTOGRAPHIC PROCESS?

Joseph Niépce and Louis J. M. Daguerre. Niépce was successful not only in getting pictures produced in the camera, but he succeeded in “fixing” them permanently, Daguerre developed a process known as “daguerreotype,” which was the first method of photography available for practical purposes. This was in 1837. With the general acceptance of daguerreotypes, photography became a profession. The process had no rival until about 1851, when the “collodion process” was discovered, and, after that, the daguerreotype process became obsolete.

WHO DEVELOPED THE MODERN PROCESS OF PHOTOGRAPHY?

William Henry Fox Talbot, an English inventor (1800-1877). He greatly increased the sensitiveness of paper, and from his negatives prints were produced in much the same way as in the present day.

THE MENTOR

The Wealth of The Mentor

1 Beautiful Children in Art, by Gustav Kobbé.
2 Makers of American Poetry, by Hamilton Mabie
3 Washington the Capital, by Dwight Elmendorf.
4 Beautiful Women in Art, by Willing.
5 Romantic Ireland, by Elmendorf.
6 Masters of Music, by Henderson.
7 Natural Wonders of America, by Elmendorf.
8 Pictures We Love to Live With, by Huneker.
9 The Conquest of the Peaks, by Fay.
10 Scotland, the Land of Song and Scenery, by Elmendorf.
11 Cherubs in Art, by Kobbé.
12 Statues with a Story, by Lorado Taft.
13 The Discoverers, by Albert Bushnell Hart.
14 London, by Elmendorf.
15 The Story of Panama, by Bonsal.
16 American Birds of Beauty, by E. H. Forbush.
17 Dutch Masterpieces, by Van Dyke.
18 Paris, the Incomparable, by Elmendorf.
19 Flowers of Decoration, by H. S. Adams.
20 Makers of American Humor, by Burges Johnson.
21 American Sea Painters, by Arthur Hoeber.
22 The Explorers, by Albert Bushnell Hart.
23 Sporting Vacations, by Dan Beard.
24 Switzerland, the Land of Scenic Splendors, by Elmendorf.
25 American Novelists, by Hamilton Mabie.
26 American Landscape Painters, by Samuel Isham.
27 Venice, the Island City, by Elmendorf.
28 The Wife in Art, by Kobbé.
29 Great American Inventors, by Bruce.
30 Furniture and Its Makers, by Richards.
31 Spain and Gibraltar, by Elmendorf.
32 Historic Spots of America, by McElroy.

33 Beautiful Buildings of the World, by Ward.
34 Game Birds of America, by E. H. Forbush.
35 The Contest for North America, by A. B. Hart.
36 Famous American Sculptors, by Lorado Taft.
37 The Conquest of the Poles, by Rear Admiral Peary.
38 Napoleon, by Ida M. Tarbell.
39 The Mediterranean, by Elmendorf.
40 Angels in Art, by Van Dyke.
41 Famous Composers, by Henry T. Finck.
42 Egypt, the Land of Mystery, by Elmendorf.
43 The Revolution, by Albert Bushnell Hart.
44 Famous English Poets, by Mabie.
45 Makers of American Art, by J. T. Willing.
46 The Ruins of Rome, by Botsford.
47 Makers of Modern Opera, by H. E. Krehbiel.
48 Two Early German Painters—Dürer and Holbein, by F. J. Mather, Jr.
49 Vienna, the Queen City, by Elmendorf.
50 Ancient Athens, by Botsford.
51 The Barbizon School, by Hoeber.
52 Abraham Lincoln, by A. B. Hart.
53 George Washington, by McElroy.
54 Mexico, by Frederick Palmer.
55 Famous American Women Painters, by Arthur Hoeber.
56 The Conquest of the Air, by Woodhouse.
57 Court Painters of France, by Coffin, N. A.
58 Holland, by Elmendorf.
59 Our Feathered Friends, by E. H. Forbush.
60 Glacier National Park, by Hornaday.
61 Michelangelo, by Cox.
62 American Colonial Furniture, by Esther Singleton.
63 American Wild Flowers, by Eaton.
64 Gothic Architecture, by Ward.
65 The Story of the Rhine, by Elmendorf.
66 Shakespeare, by Mabie.
67 American Mural Painters, by Hoeber.
68 Celebrated Animal Characters, by Hornaday.

69 Japan, by Elmendorf.
70 The Story of the French Revolution, by Albert Bushnell Hart.
71 Rugs and Rug Making, by Mumford.
72 Alaska, by Browne.
73 Charles Dickens, by Mabie.
74 Grecian Masterpieces, by Lorado Taft.
75 Fathers of the Constitution, by Albert Bushnell Hart.
76 Masters of the Piano, by Finck.
77 American Historic Homes, by Singleton.
78 Beauty Spots of India, by Elmendorf.
79 Etchers and Etching, by Weitenkampf.
80 Oliver Cromwell, by Albert Bushnell Hart.
81 China, by Elmendorf.
82 Favorite Trees, by Hornaday.
83 Yellowstone National Park, by Elmendorf.
84 Famous Women Writers of England, by Mabie.
85 Painters of Western Life, by Hoeber.
86 China and Pottery of Our Forefathers, by Esther Singleton.
87 The Story of The American Railroad, by Albert Bushnell Hart.
88 Butterflies, by Holland.
89 The Philippine Islands, by Worcester.
90 Great Galleries of the World—the Louvre, by Van Dyke.
91 William M. Thackeray, by Mabie.
92 The Grand Canyon, by Elmendorf.
93 Architecture in American Country Homes, by Aymar Embury.
94 The Story of the Danube, by Albert Bushnell Hart.
95 Animals in Art, by Kobbé.
96 The Holy Land, by Elmendorf.
97 John Milton, by Mabie.
98 Joan of Arc, by Ida M. Tarbell.
99 Furniture of the Revolutionary Period, by Esther Singleton.
100 The Ring of the Nibelung, by Finck.
101 The Golden Age of Greece, by Botsford.
102 Chinese Rugs, by Mumford.
103 The War of 1812, by Albert Bushnell Hart.
104 Great Galleries of the World—The National Gallery, London, by Van Dyke.

105 Masters of the Violin, by Finck.
106 American Pioneer Prose Writers, by Mabie.
107 Old Silver, by Esther Singleton.
108 Shakespeare’s Country, by William Winter.
109 Historic Gardens of New England, by Mary H. Northend.
110 The Weather, by C. F. Talman.
111 American Poets of the Soil, by Johnson.
112 Argentina, Newman.
113 Game Animals of America, by Hornaday.
114 Raphael, by Van Dyke.
115 Walter Scott, by Mabie.
116 The Yosemite Valley, by Elmendorf.
117 John Paul Jones, by Albert Bushnell Hart.
118 Russian Music, by Finck.
119 Chile, by Newman.
120 Rembrandt, by Van Dyke.
121 Southern California, by C. F. Lummis.
122 Keeping Time, by Talman.
123 American Miniature Painting, by Mrs. Elizabeth Lounsbery.
124 Gems, by Esther Singleton.
125 The Orchestra, by Henderson.
126 Brazil, by E. M. Newman.
127 The American Triumvirate, by A. B. Hart.
128 The Madonna and Child in Art, by Van Dyke.
129 The Story of the American Navy, by Barnes.
130 Lace and Lace Making, by Esther Singleton.
131 American Water Color Painters, by Kobbé.
132 Peru, by E. M. Newman.
133 The Story of the American Army, by Hart.
134 Our Planet Neighbors, by Harold Jacoby.
135 The Story of Russia, by Leo Pasvolsky.
136 The Story of the Hudson, by A. B. Hart.
137 Prehistoric Animal Life, by Dr. Matthew.
138 Hawaii, by E. M. Newman.
139 Earthquakes and Volcanoes, by Talman.
140 The Canadian Rockies, by Ruth Kedzie Wood.
141 Corot, by Elliott Daingerfield.

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